

Student Assignment Brief

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Assignment Information

Module Name:	Database System
Module Code:	ST4005CEM
Assignment Title:	ST4005CEM Database System March Intake 2026 Coursework CW WITH VIVA (Regular)
Assignment Due:	Jul 29, 2026

Assignment Credit:	5
Word Count:	2000
Assignment Type:	Coursework : Individual Report
Grading:	Percentage Grade(Applied Core Assessment).

Assessment Overview

You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

Important Notice

The work you submit for this assignment must be your **own independent work**. More information is available in the 'Assignment Task' section of this assignment brief.

Assessed Module Learning Outcomes

The Learning Outcomes for this module align with the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task. The following module learning outcomes are assessed in this task

Assignment Task

Assignment Task

Introduction

This assignment contains THREE (3) parts: design, data and queries, and analysis. All parts of the assessment relate to the "BuildCorp Infrastructure Equipment Logistics" scenario outlined below.

Background and Requirements

BuildCorp International is a rapidly expanding engineering firm based in Kathmandu, managing large-scale physical infrastructure projects across Nepal and internationally. Currently, the company uses fragmented spreadsheets to manage project lifecycles, equipment hire, and HR coordination. This leads to significant data redundancy, inconsistencies in billing, and difficulty in retrieving useful insights about project timelines or equipment availability.

The Chief Operations Officer, Ms. Aasha Thapa, has proposed a centralized relational database system to manage projects, equipment inventory, hire rates, and personnel assignments. The

system aims to reduce redundancy, ensure secure record-keeping, and support future analytics needs.

Scenario

BuildCorp manages various Projects, each with a specific name, owner, and defined start/end dates. Every project is assigned a primary Contact person categorized by Type (e.g, Client, Contractor).

The core operations involve Equipment management. Equipment is classified into multiple Categories (e.g, Heavy Machinery, Safety Gear), and each item has a Status (Active, Maintenance). Each piece of equipment has specific Hire Rates that vary by period (Daily/Weekly).

Logistically, the system tracks Calendar entries linking projects, equipment, and confirmed invoice dates. Furthermore, Project HR assignments track specific personnel testimonials for each project, while Project Equipment logs the specific booking rates for every piece of machinery used

The "Messy" Data

Students must use these tables as a starting point for normalization, ensuring the design reaches Third Normal Form (3NF)

Table 1: Project Master Contact Registry

Proj_ID	Project_Name_&Owner	Contact_Person_Details	Contact_Type	Proj_Dates
P-901	Bagmati Bridge (Govt. Nepal)	Rajesh Hamal (r.hamal@email.com)	Lead Engineer	2026-01-01 to 2027-12-31
P-902	Pokhara Airport (Civil Aviation)	Sita Rai (sita.r@email.com)	Site Manager	2026-02-15 to 2028-06-30
P-903	Kathmandu Metro (KMC)	Binod Das (b.das@email.com)	Lead Engineer	2026-03-01 to 2029-01-01

Table 2: Equipment Catalog Multi-Categories

Equip_ID	Machine_Name	Category_List	Current_Status	Status_Notes
E-101	Caterpillar Excavator 320	Heavy, Earthmoving, Hydraulic	Active	Fully Operational
E-102	Tower Crane XJ	Heavy, Lifting	Maintenance	Returning March 2026
E-103	Volvo Haul Truck	Heavy, Transport	Active	New Tires Installed

Table 3: Complex Hire Rate Tiers

Rate_ID	Equip_Linked	Daily_Rate	Weekly_Rate	Hire_Period_Notes
HR-55	E-101	150	900	Includes Operator Fuel
HR-60	E-102	500	3000	Dry Hire Only
HR-65	E-103	250	1400	Weekly discount applied

Table 4: Global Contact Type Mapping

Type_ID	Type_Name	Security_Level	Dept_Head
T-1	Lead Engineer	Level 4	Dr. Binod
T-2	Site Manager	Level 3	Er. Suman
T-3	Client Rep	Level 2	Ms. Thapa

Table 5: Project HR Testimonials

HR_Rec_ID	Project_ID	HR_Name_&Role	Testimonial_Text
REC-01	P-901	Rajesh Hamal (Lead Engineer)	"Excellent hydraulic power on site."
REC-02	P-902	Sita Rai (Site Manager)	"Crane maintenance was slow."
REC-03	P-901	Binod Das (Consultant)	"Project ahead of schedule."

Table 6: Equipment Category Bridge (Raw Log)

Bridge_ID	Equipment_ID	Category_ID_Mapping	Last_Audit_Date
B-1	E-101	CAT-01, CAT-04, CAT-07	2025-12-01
B-2	E-102	CAT-01, CAT-05	2025-11-15
B-3	E-103	CAT-01, CAT-09	2026-01-10

Table 7: Master Booking Calendar

Cal_ID	Booking_Date	Project_ID	Equip_ID	Invoice_Status
C-1001	2026-02-15	P-901	E-101	Confirmed: INV-442
C-1002	2026-02-16	P-901	E-101	Pending: NO_INV
C-1003	2026-02-17	P-902	E-103	Confirmed: INV-445

Table 8: Project Equipment Financials

Proj_Equip_ID	Project_ID	Equip_ID	Booking_Rate_Applied	Duration_Days
PE-55	P-901	E-101	145	10
PE-56	P-902	E-103	250	5
PE-57	P-901	E-103	230	15

Please note: The data shown in the assignment is not necessarily normalized, and it is the candidate's task to organize the data in the most optimal way possible.

For example:

The records shown below do not map directly to database tables

There may be repeating data

Not all values are atomic

Not all unique identifiers are present or clearly defined

The candidate is expected to use these tables as a starting point for normalization, ensuring the design reaches Third Normal Form (3NF) or higher.

Please note also that all SQL scripts should be shown along with their results.

1. Create Tables (5 Marks)

Using your normalized schema and the provided ER diagram:

Write SQL CREATE TABLE statements for each entity (e.g, tblPROJECTS, tblEQUIPMENT, tblCALENDAR).

Include Primary Keys, Foreign Keys, and appropriate data types (VARCHAR, DATE, DECIMAL).

Apply constraints such as NOT NULL, UNIQUE, CHECK, and DEFAULT.

Output: Show confirmation of successful table creation using DESCRIBE table_name.

2. Insert Sample Data (5 Marks)

Insert realistic sample data into each table using INSERT INTO statements.

Populate each table with at least 35 sample records that reflect the structure of the 8 unnormalized legacy tables.

Output: Show outputs using SELECT * FROM table_name.

Constraint: Ensure that foreign key relationships are respected in the order of insertion.

3. Write SQL Queries (15 Marks)

Write and execute the following SQL queries based on your database:

Average Hire Rates: Calculate the average hire_rate_rate across all equipment categories. (1 Mark)

Contact Directory: Display the names and emails of all contacts in alphabetical order. (1 Mark)

Active Resource Check: Show all project IDs and start dates for projects that have booked more than one piece of equipment in the tblCALENDAR. (1 Mark)

Unique Statuses: Count the number of unique equipment statuses (e.g, Active, Maintenance) available in tblSTATUS. (1 Mark)

Targeted Search: List all equipment names that include the term "Excavator". (1 Mark)

Owner Update: Update the phone number of the contact person "Rajesh Hamal" to "9800000000". (1 Mark)

Inventory Expansion: Add a new equipment item "JCB Backhoe" with sample details to the tblEQUIPMENT catalog. (1 Mark)

Data Cleanup: Delete the record in tblPROJECT_EQUIPMENT that refers to an equipment ID with a booking rate of 0. (1 Mark)

Allocation Summary: Show the total number of equipment items assigned to each project using GROUP BY. (2 Marks)

Project Audit: Retrieve all equipment names and booking rates associated with a given project_id. (2 Marks)

Client Engagement: Count the total number of projects associated with each primary contact. (1 Mark)

Idle Inventory: Display a list of equipment that has not been assigned to any project calendar yet. (1 Mark)

System Upgrade: Add a new column Status to the tblPROJECTS table and set all existing records to "Active". (1 Mark)

Task 3 - Data Visualization (25 Marks)

You are required to visualize the different graphs like Line graphs, Bar graphs, hist graphs, and Pie charts and analyse it from the given dataset below.

Dataset link: [**Dataset For Visualization**](#)

Submission Instructions

Requirement	Details
File Naming	NAME_studentID
File Format	.docx/.pdf format
Submission Method	Campus 4.0 platform (submission link provided 2 weeks before deadline)

Marking and Feedback

How will my assignment be marked?

Your assignment will be marked by the Module Team using standardized criteria.

How will I receive grades and feedback?

Provisional marks will be released once internally moderated. Feedback will be provided alongside grades release within 2 weeks (10 working days).

What will I be marked against?

Details of the marking criteria for this task can be found in the Assessment Marking Criteria section at the end of this brief.

Grade Requirements

You must achieve 40% or above to pass this assessment. Ensure you understand the marking criteria for successful completion.

Assignment Support and Academic Integrity

Getting Help

If you have any questions about this assignment, please meet your respective module leader or teacher for more information.

Language Standards

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity

The work you submit must be your own. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work, **excluding Artificial Intelligence (AI)**.

We use detection software and make routine checks for evidence of academic misconduct. Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found in Student handbook in Campus 4.0.

All cases of suspected academic misconduct are referred to for investigation, the outcomes of which can have profound consequences to your studies.

Support for Students with Disabilities

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms, contact the Student Support Office for assistance.

Unable to Submit on Time?

If events prevent you from submitting on time, guidance on extenuating circumstances is available in the Student Handbook or from the Student Support Office.

Administration of Assessment

Module Leader Name:

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Module Leader Email:

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Attempt Type:	regular
Component Code:	cw with viva

Assessment Marking Criteria

T1: Normalization (10%)

0	1	2	3	4	5
Failed to submit the task within the deadline. Normalization is not done.	Only up to 1NF is shown or explained. Single value attribute is maintained.	Complete 1NF is done. Dependency is identified and also removed.	Up to 2NF is shown or explained. Partial dependency is identified and removed.	Up to 3NF is shown or explained. Transitive or non-key dependency is identified and removed.	Excellent Normalization. All functional dependency is identified and removed. Shown in tabular format with clear model and all relevant relation and attributes.

T2: Entity Relationship Diagram (15%)

0	1	2	3	4	5
No ER diagram. Process of normalization is not documented.	ER diagram with only few relevant entities. Process of normalization is not documented.	ER diagram with relevant entities and their attributes, their type, and depicted. Process of normalization is documented.	Good ER diagram with relevant entities and their cardinalities identified and depicted. Process of normalization is documented.	Good ER diagram with relevant entities and their cardinalities identified and depicted. Also, justifiably identified.	Excellent ER diagram with relevant entities, their cardinalities, primary key, foreign key identified and depicted. Process of normalization is documented.

T3: SQL (25%)

0	1	2	3	4	5
No SQL query.	Only CREATE TABLE query without keys and proper datatype.	CREATE TABLE query with keys and relevant attribute with proper datatype.	CREATE TABLE query with keys and relevant attribute, proper datatype and length of attribute.	CREATE TABLE query with keys, relevant attribute, proper datatype and length of attribute along with output. Insert query with significant data.	CREATE TABLE query with keys, relevant attribute, proper datatype and length of attribute along with output. Insert query with significant data. Shown one SELECT and one JOIN query.

T4: Data Visualization (25%)

0	1	2	3	4	5
No visualization.	Only import of sample data into modeling tool.	Performing linear regression plot along with correlation.	Performing linear regression plot along with correlation and fit the linear regression.	Performing linear regression plot along with correlation and fit the linear regression.	Summary of the model along with correlation and fit of linear regression.

T5: Viva (25%)

0	1	2	3	4	5
Viva not given.	Only a few tasks explained in viva session.	Only a few tasks explained in viva session with little basic concept.	Good summary of completed task in assignment with how and what done.	Excellent summary of completed task in assignment with how and what done.	Excellent summary of completed task in assignment with how and what done, proper output, and justification.

Task1: Normalization (10%)

Reflect understanding of dependencies and normalization.

Task 2: Entity Relationship diagram (15%)

Draw an entity relationship diagram for this system. Identify all relevant entities, attributes, suitable attribute types, primary keys, foreign keys, and cardinalities between entities. This should be normalized up to the 3rd Normal Form. You should document all the steps you took to come up with this normalized ER diagram.

Task 3: SQL (25%)

Create a database using MySQL- with primary keys, foreign keys, and other attributes mentioned for each entity using proper constraints. Input some significant data in each table showing your understanding of the scenario. Write one SELECT and one JOIN query

This document is confidential and intended solely for enrolled students.